Penetration Test - Executive Summary

This authorized penetration test evaluated a simulated web application, file-upload functionality, an exposed network echo service, and post-exploitation controls in an isolated lab environment. The primary exploitation scenario involved abusing insecure file-upload handling to place a PHP reverse-shell payload on the web server and establish a test connection through a Metasploit handler, demonstrating the potential for remote command execution. Network fuzzing also highlighted how an exposed echo service could be misused for reconnaissance or unexpected-input testing, while a user-crontab persistence exercise illustrated the risk of continued unauthorized access after initial compromise. The main risk is that unrestricted executable uploads could permit an external attacker to compromise system confidentiality, integrity, and availability. Recommended mitigations include strict upload allowlisting, storage outside executable web directories, least-privilege service accounts, egress controls, centralized logging, and alerts for suspicious web processes, outbound connections, and cron changes. All findings and activity described are from simulated, authorized testing; they do not establish that a production system was compromised.